The Cixiang Road Big Two Championship

Rules, Results, and an Exploratory Statistical Analysis

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1 Background

This competition began with a series of increasingly spirited, and arguably unnecessary, arguments between Popo and Flower about who was better at cards. To give the dispute a formal and reasonably fair resolution, Terry and Teddy joined the contest, turning it into a four-player championship with recorded scores.

This report documents all 20 rounds of the Cixiang Road Big Two Championship, including individual round scores, subtotals for two blocks of ten rounds, and final standings. An exploratory statistical analysis assesses whether the players differ in their mean net scores per round.

The distinction is important: identifying the winner of these 20 rounds is not the same as establishing who performs best in the long run.

2 Game Rules and Scoring

2.1 Overview

The competition used a four-player Taiwanese house-rule version of Big Two. A standard 52-card deck was used without jokers, and each player received 13 cards.

The objective was to be the first player to empty their hand. Players took turns playing cards or legal combinations, subject to the response rules described below. The first player to use all their cards won the round, which ended immediately.

The remaining three players lost points according to the cards left in their hands and the applicable penalties. The winner received the sum of all three opponents' losses.

The following rules describe this group's version of the game; they should not be interpreted as universal Big Two rules.

2.2 Dealing Procedure and Direction of Play

After shuffling, the deck was opened at an interior point to reveal a card. The card was inspected without being removed, and the deck was closed again before dealing.

For this procedure, A counted as 1, J as 11, Q as 12, and K as 13. Numbered cards retained their numerical values. These numerical values were used to determine the starting recipient of the deal.

The revealed card's value was divided by four. The dealer was designated as Player 1 and chose whether to count seats clockwise or counterclockwise. Remainders of 1, 2, and 3 indicated Players 1, 2, and 3 in the chosen counting direction, respectively. A remainder of 0 indicated Player 4.

For example, revealing a 6 gave a remainder of 2, so the first card was dealt to Player 2.

This procedure determined the starting recipient of the deal, not the player making the opening play.

The direction of play was opposite to the direction of dealing:

- Clockwise dealing was followed by counterclockwise play.
- Counterclockwise dealing was followed by clockwise play.

2.3 Opening Play

The player holding the three of clubs made the opening play of each round.

The opening play had to include the three of clubs, but was not restricted to a single card. Any otherwise legal combination containing the three of clubs could be used as the opening play.

2.4 Turn Sequence and Ordinary Plays

The player leading a sequence could choose a legal play. Subsequent players took turns either beating the current play or passing.

Except for the bomb exception described below, a response had to match the current combination type and contain the same number of cards. A single card therefore required a higher single, a pair required a higher pair, and a straight required a higher straight.

Having the same number of cards was not sufficient: a full house could not be played over a straight simply because both contained five cards.

The ordinary legal plays were:

- **Single:** one card.
- **Pair:** two cards of the same rank.
- **Straight:** five cards forming a permitted consecutive-rank sequence.
- **Full house:** three cards of one rank together with two cards of another rank.

Two-pair combinations, standalone three-of-a-kind (trips), and flushes that were not straight flushes were not legal plays under these house rules. Trips was permitted only as part of a full house.

2.5 Straight and Suit Rankings

Under these house rules, A2345 was the lowest straight and 23456 was the highest.

The intervening ordinary sequences followed increasing rank, giving the following order from lowest to highest:

$$A2345 < 34567 < 45678 < 56789 < 6789\,10 < 789\,10J < 89\,10JQ < 9\,10JQK < 10JQKA < 23456.$$

The suit ranking, from highest to lowest, was:

$$\text{Spades} > \text{Hearts} > \text{Diamonds} > \text{Clubs}.$$

These rankings describe the house rules of this competition.

2.6 Special Plays: Quads and Straight Flushes

Quads and straight flushes were treated as special plays, or "bombs."

A quads play consisted of four cards of the same rank plus one additional card, for a total of five cards. Quads were compared by the rank of the four matching cards; the additional card did not determine their ranking. For example, four 8s with an additional card beat four 7s with an additional card.

A straight flush consisted of five cards forming a permitted straight sequence in the same suit.

On a player's turn, a bomb could beat an ordinary play without matching its combination type or card count. For example, a bomb could be played against a single card, a pair, a straight, or a full house. This exception did not permit interrupting another player's turn.

Any straight flush ranked above any quads play. Straight flushes were first compared by their straight sequence, using the rankings described above: A2345 was the lowest and 23456 the highest. For example, a 34567 straight flush beat an A2345 straight flush.

If two straight flushes had the same rank sequence, their suits determined the winner. For example, a spade straight flush beat a heart straight flush with the same rank sequence.

A bomb could therefore be beaten only by a higher-ranking bomb, not by an ordinary play.

2.7 Ending a Sequence and a Round

When the other three players passed consecutively, the player who made the last successful play led a new sequence and could choose a new legal play type.

The first player to empty their hand won the round. Play then stopped immediately, and the remaining hands were scored.

2.8 Base Scores and Doubling Penalties

Each remaining card counted as one penalty point, subject to the following multipliers:

- **Remaining-card penalty:** A player with more than 10 cards remaining, meaning 11 to 13 cards, had their loss doubled. Exactly 10 remaining cards did not trigger this penalty.
- **Unplayed-2 penalty:** Each card of rank 2 remaining in a player's hand doubled their loss again. One remaining 2 produced a multiplier of 2; two produced a multiplier of 4; three produced a multiplier of 8; and four produced a multiplier of 16.
- **Combined penalties:** The remaining-card penalty and the unplayed-2 penalties were applied multiplicatively.

For a losing player with r cards remaining, including k cards of rank 2, the point loss L was defined as:

$$L = \begin{cases} r \times 2^{k}, & 1 \le r \le 10, \\ r \times 2^{k+1}, & 11 \le r \le 13. \end{cases}$$

For example, six remaining cards including one 2 resulted in a loss of:

$$6 \times 2 = 12 \text{ points.}$$

Eleven remaining cards including two 2s resulted in a loss of:

$$11 \times 2 \times 4 = 88 \text{ points.}$$

2.9 Net Score per Round

Each losing player's net score was recorded as $-L$. The winner's net score equaled the sum of the other three players' point losses. Consequently, the four net scores summed to zero in every round.

All analyses used these final net scores, including the applicable doubling penalties.

Only final net scores were recorded. Remaining-card counts and the number of unplayed 2s were not documented for individual rounds. The contributions of the different penalties therefore cannot be fully reconstructed from the available data.

3 Recorded Results

Table 1 presents the net scores for all 20 rounds. Positive values indicate points gained, and negative values indicate points lost. All scores include the applicable penalties.

The four scores sum to zero in each round. The first and second blocks of ten rounds follow the order of the two original score sheets; they are not assumed to represent different dates or independent playing sessions.

Table 1: Complete scores for the 20-round championship

Round	**Flower**	**Popo**	**Terry**	**Teddy**
1	−1	−5	−2	+8
2	−2	+6	−3	−1
3	−1	−1	−2	+4
4	−5	−9	+17	−3
5	−14*	−22*	+45	−9
6	+15	−3	−6	−6
7	−6	−20*	−6	+32
8	−22*	−9	+51	−20*
9	−12*	+21	−6	−3
10	−1	−2	−9	+12
Rounds 1–10 subtotal	−49	−44	+79	+14
11	+10	−1	−7	−2
12	−1	+7	−5	−1
13	−6	+11	−1	−4
14	−6	+23	−9	−8
15	+33	−5	−20*	−8
16	+12	−7	−4	−1
17	−6	+30	−20*	−4
18	−4	−22*	−2	+28
19	−2	+9	−4	−3
20	−2	−2	+7	−3
Rounds 11–20 subtotal	+28	+43	−65	−6
Total net score	**−21**	**−1**	**+14**	**+8**
Rounds won	4	7	4	5

Note: * indicates a net loss exceeding nine points, classified as a penalty event under the competition's scoring convention. All scores include applicable doubling penalties.

4 Statistical Methods

4.1 Objective and Outcome Measures

The primary objective was to compare the players' long-run mean net scores per round under the scoring rules and opponent composition of this competition.

The primary outcome was the final net score per round, including all doubling penalties. The number of rounds won was reported descriptively and was not the primary basis for comparison.

The mean net score per round for player j was defined as:

$$\bar{x}_j = \frac{1}{20} \sum_{i=1}^{20} x_{ij},$$

where x_{ij} denotes player j's net score in round i.

Descriptive statistics included the total score, mean score per round, sample standard deviation of round scores, and a 95% confidence interval for the mean.

4.2 Paired Comparisons and Multiple-Testing Correction

The four scores within a round were related. The dataset was therefore treated as 20 sets of paired observations, rather than 80 independent observations.

All six pairwise comparisons were evaluated using two-sided paired t-tests [1]. No single player was designated as the reference.

For any two players A and B, the round-specific difference was calculated as:

$$d_i = x_{iA} - x_{iB}.$$

The null hypothesis was that their long-run mean difference was zero. The test statistic was:

$$t = \frac{\bar{d}}{s_d/\sqrt{20}},$$

where $\bar{d}$ is the sample mean difference and s_d is the sample standard deviation of the round-specific differences. Each test had 19 degrees of freedom.

The six p-values were adjusted using Holm's procedure [2] to control the family-wise Type I error rate at $\alpha = 0.05$.

All 95% confidence intervals were calculated using the t distribution. These intervals were not adjusted for multiple comparisons.

4.3 Analytical Assumptions and Sensitivity Analysis

The primary analysis assumed approximately independent rounds and paired score differences suitable for t-based inference. Given the small sample and occasional large scores, all inferential results were considered exploratory.

As a sensitivity analysis, exact two-sided paired sign-flip tests were conducted for all six comparisons. For each pair, all 2^{20} possible sign assignments of the 20 observed score differences were enumerated, using the absolute mean difference as the test statistic. The resulting p-values were adjusted using Holm's procedure.

Exact sign-flip inference requires the joint distribution of paired differences to be invariant under the sign changes used. In particular, symmetry and independence are sufficient conditions; the procedure does not automatically correct for serial dependence between rounds. All scores, including doubling penalties, were retained. Only round-level net scores, rather than cumulative totals, were treated as observations.

5 Statistical Results

5.1 Player Score Distributions

Terry finished first with a total of +14 points, followed by Teddy (+8), Popo (−1), and Flower (−21).

Their respective mean net scores per round were +0.70, +0.40, −0.05, and −1.05 points (Table 2).

Terry had the largest standard deviation of round scores (17.99 points), while Flower had the smallest (11.61 points).

These values describe the variability observed in this competition. They do not directly establish the quality of a player's strategy or their long-term consistency.

Table 2: Descriptive statistics for net scores across 20 rounds

Player	Total	Mean per round	SD	95% CI for mean
Terry	+14	+0.70	17.99	[−7.72, 9.12]
Teddy	+8	+0.40	12.03	[−5.23, 6.03]
Popo	−1	−0.05	14.15	[−6.67, 6.57]
Flower	−21	−1.05	11.61	[−6.48, 4.38]

Note: SD, sample standard deviation; CI, confidence interval. Means, SDs, and confidence intervals are expressed in points per round; totals are expressed in points. Confidence intervals were not adjusted for multiple comparisons.

5.2 Six Pairwise Comparisons

None of the pairwise comparisons reached statistical significance. Unadjusted p-values ranged from 0.701 to 0.958, and all Holm-adjusted p-values were 1.000 (Table 3).

The largest observed absolute mean difference was between Terry and Flower: Terry averaged 1.75 more points per round. However, the unadjusted p-value was 0.772 and the Holm-adjusted p-value was 1.000.

Terry averaged only 0.30 more points per round than Teddy. The 95% confidence interval for this difference ranged from −11.39 to 11.99 points per round, indicating substantial uncertainty.

Table 3: Pairwise comparisons of net scores per round

Comparison	Mean difference	95% CI	Raw p	Holm p
Flower–Popo	−1.00	[−9.84, 7.84]	0.815	1.000
Flower–Terry	−1.75	[−14.21, 10.71]	0.772	1.000
Flower–Teddy	−1.45	[−9.24, 6.34]	0.701	1.000
Popo–Terry	−0.75	[−13.71, 12.21]	0.905	1.000
Popo–Teddy	−0.45	[−10.72, 9.82]	0.928	1.000
Terry–Teddy	+0.30	[−11.39, 11.99]	0.958	1.000

Note: Differences are calculated as the first player minus the second, in points per round. Tests were two-sided paired t-tests. Holm adjustment included all six comparisons. Confidence intervals were not adjusted for multiple comparisons.

Sensitivity Analysis. Exact paired sign-flip tests yielded unadjusted p-values of 0.824, 0.802, 0.718, 0.913, 0.937, and 0.966, respectively, in the order shown in Table 3. All six Holm-adjusted p-values were 1.000. Neither analysis detected a significant pairwise difference; this agreement does not validate their assumptions or demonstrate equivalent player ability.

6 Discussion

6.1 Individual Performances

- **Terry (first place, +14 points):** Terry accumulated substantial gains in the first ten rounds, particularly in Round 5 (+45) and Round 8 (+51), finishing that block with a net score of +79. Despite losing 65 points over the next ten rounds, the earlier gains were sufficient to secure first place.
- **Teddy (second place, +8 points):** Teddy gained 14 points in the first ten rounds and gave back six in the second block. He finished positive and six points behind the winner. His round-score variability was lower than Terry's and Popo's, but was not the lowest of the four.
- **Popo (third place, −1 point):** Popo won seven rounds, more than any other player. However, substantial losses in some rounds left the final score slightly negative. This illustrates that winning the most rounds does not necessarily produce the highest total score.
- **Flower (fourth place, −21 points):** Flower lost 49 points in the first ten rounds and recovered 28 in the second block. The +33 score in Round 15 was the largest single-round gain in the final ten rounds, but the recovery did not fully offset the earlier losses.

6.2 Event Standings Versus Long-Term Performance

The standings of this competition can be determined directly from the total scores. They should not, however, be treated as an established ranking of long-term playing ability.

For example, Terry scored +79 and −65 in the two blocks of ten rounds, illustrating the considerable variation possible within a short series of games.

None of the six comparisons was statistically significant. The available data therefore do not provide sufficient evidence of a difference in long-run mean scores between any pair of players.

This does not establish that the players are equally skilled. It also does not establish that the results were entirely determined by luck.

6.3 Limitations and Future Data Collection

The analysis has several limitations:

- **Small sample:** Only 20 rounds were available. Confidence intervals for mean differences were wide, limiting the ability to identify modest long-term advantages.
- **Statistical assumptions:** Extreme scores and the small sample may affect t-based inference. Exact sign-flip tests provide a sensitivity check, but require sign-exchangeability. Neither method addresses potential serial dependence caused by fatigue, seating, or carry-over effects.
- **Missing penalty and gameplay records:** Remaining-card counts, unplayed 2s, and the reasons for individual penalties were not recorded. The contributions of different penalties cannot be separated, and the effectiveness of particular playing strategies cannot be determined.
- **Scope of the rules description:** The report documents the dealing procedure, opening-play requirement, permitted combinations, bomb hierarchy, suit ranking, and scoring system. Some ordinary-play ranking and tie-breaking details are not specified.

- **Limited generalizability:** The results describe these four players under this opponent composition and scoring system.

The present 20 rounds can serve as pilot data for planning a future study. A meaningful target difference, sample size, and stopping rule should be specified in advance.

Future games should use fixed, fully documented rules, thorough shuffling, and a predetermined seating-rotation schedule. Net scores, dates, and session identifiers should be recorded for every round. Data collection should not stop simply because a statistically significant result has appeared.

If rounds within a session are correlated, the analysis should account for session-level clustering, rather than treating every round as fully independent.

6.4 Planning the Number of Future Rounds

Estimating a suitable sample size requires specifying the difference that the study aims to detect.

For illustration, this planning exercise uses a true mean difference of three points per round. It does not assume that the differences observed in this small pilot equal the true long-run differences.

Across the six pairs, the sample standard deviations of round-specific score differences were approximately 17 to 28 points.

For sample-size planning, the paired score differences were treated as a single outcome whose mean would be compared with zero. A normal-approximation calculation was used for this one-sample problem [3].

Assuming a family-wise significance level of 0.05, 80% power, and a conservative Bonferroni threshold for six two-sided comparisons [2], the required number of rounds can be approximated by:

$$n \approx \left(\frac{\left(z_{1-0.05/(2\times 6)} + z_{0.80} \right) s_d}{3} \right)^2,$$

where s_d is the standard deviation of round-specific paired differences and z_q is the qth quantile of the standard normal distribution.

Holm adjustment can still be used in the final analysis.

Using the unrounded pilot standard deviations, the normal approximation gives approximately 373 to 1,032 rounds across the six comparisons. For practical planning, this is rounded upward to approximately 400 to 1,100 rounds. The required sample size differs between pairs because their score-difference variabilities differ.

To use a common sample size that accommodates the more variable pairs, approximately 1,100 new rounds could serve as an initial planning target, with the current 20 rounds retained as pilot data.

Here, 80% power refers to the approximate probability of detecting a difference for a particular pair whose true mean scores differ by three points per round, under the stated assumptions.

It does not guarantee a significant result after the planned number of rounds. Nor does it provide 80% power to distinguish all six pairs simultaneously or establish a complete ranking of the four players. Smaller true differences generally require more rounds.

This estimate assumes approximate independence between rounds and future variability similar to that observed in the pilot. Because the pilot contains only 20 rounds, the variability estimates and resulting sample-size estimates remain uncertain.

Within-session correlation or substantial changes in player performance over time would require further consideration in the study design and sample-size calculation.

7 Conclusion

Terry won the 20-round Cixiang Road Big Two Championship with a total score of +14 points. Teddy, Popo, and Flower followed with +8, −1, and −21 points, respectively.

None of the six paired comparisons of net scores was statistically significant after Holm adjustment. The results establish the standings of this particular competition, but do not provide sufficient evidence of a long-term mean-score advantage for any player. A lack of statistical significance should not be interpreted as proof of equal ability.

To detect a true mean difference of three points per round with approximately 80% power for an individual pair, the observed variability suggests a rough requirement of 400 to 1,100 new rounds under the stated assumptions. This is a conditional planning estimate, not a universal threshold for determining who is best.

Official ruling: Terry may claim this championship. Who gets to rule Cixiang Road in the long run remains a question for future games.

References

[1] NIST/SEMATECH. *e-Handbook of Statistical Methods*. Section 7.3.1.1: Analysis of paired observations. Available at: `https://www.itl.nist.gov/div898/handbook/prc/section3/prc311.htm`. Accessed October 12, 2026.

[2] Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70. Available at: `https://www.jstor.org/stable/4615733`.

[3] Brant, R. *Inference for a Mean: Comparing a Mean to a Known Value*. University of British Columbia. Available at: `https://www.stat.ubc.ca/~rollin/stats/ssize/n1.html`. Accessed October 12, 2026.